

anything about PDF as a file format. My first PDF library required that you knew how to write PDF syntax. With iText, I created an abstract layer that allowed developers to work with concepts they knew: paragraphs, anchors, lists, images, tables, cells, etc.

iText was also built for speed. As I needed to serve PDFs to a browser, I had to break some design patterns and create an architecture that was somewhat out of the ordinary, but I succeeded in making iText really fast. Over the years, several other PDF libraries emerged, but iText is still the fastest in the market.

To keep ahead of the other vendors and the copycats, we continued our investment in further development. Back in 1998, I read and followed the PDF specification. Today, iText is a member of the ISO committees for PDF and we help in writing the specification, along with companies such as Adobe, Callas, Canon, Datalogics, and many other vendors. We can make this investment because we have an open source business model that works. I don't know of any other open source PDF library that can keep up with the newest standards in the field of PDF.

Q You recently gave a lecture on intellectual property (IP). Do you think knowledge about IP is crucial for developers in the open source avenue?

When you write software — whether it's open or closed source software — you create intellectual property. So my answer is yes; knowledge about IP is crucial for developers in the open source as well as the closed source avenues. A developer can't afford to be ignorant about intellectual property. I notice that many developers assume that source code they find on the Internet can be used as if it were completely free and as if they can use it without any problem. That's a misconception.

If you find a piece of source code without a licence note, the copyright of that code is proprietary to the developer who wrote it. You can't use that source code without the explicit permission of its author. To solve this problem, we use licences. Commercial software vendors work with an End-User Licence Agreement. Open source developers add a free or open source software licence to their code. There are many different flavours of open source licences, ranging from the very liberal MIT or BSD licences to the stricter AGPL licence.

If you're an open source developer, you have to choose your open source licence wisely. For some projects, using an Apache licence is the best choice. For other projects, it's best to choose a GPL-style licence. For iText, I started with the LGPL. After a while, I added the MPL, because

people from the industry had a problem with the LGPL. In 2009, I switched to the AGPL and although some people will disagree, the AGPL is still the best licence for a project like iText.

If you're not an open source developer, but you want to use open source software, you have to know the implications of the licence that was chosen for that particular open source project. There may be specific restrictions you have to take into account if you want to use that software for free.

Q iText is already one of the successful open source enterprises. What are the factors that help monetise an open source project?

I have tried many different business models to monetise iText, starting from donations, advertisements and documentation to selling support and professional services. It took me from 2000 to 2009 to discover that none of those models worked for me. But during this decade and thanks to all these experiments, iText had become ubiquitous — every

Java and C# developer knew that iText was the library you needed to use if you had a PDF-related requirement. No developer was aware that I almost gave up on iText in 2008 because I could no longer maintain the project. My son was in hospital with cancer. I had just started my first company for iText and that company wasn't making any revenue. If I didn't find a sound business model soon, iText development was going to die a slow and painful death, in spite of its success, or maybe even because of its success. Fortunately, I found the right business model just in time. In 2009, we changed the licence from LGPL/

MPL to AGPL, making it mandatory for many companies to purchase a commercial licence if they wanted to continue using iText. In 2009, iText made its first substantial revenue and we have been growing ever since.

The most essential factor for an open source software project (any open source project) is whether people are willing to give something back for their use of the software. I find it hard to understand that some developers think that they enrich themselves by using other people's work without ever giving something back. Compare it with a parasite; a parasite lives on a host and benefits by deriving nutrients at the host's expense, sometimes killing the host in the process. To make an open source project successful, you need to create a mutual symbiotic relationship between the provider and the user. This can be a business relationship where the user becomes a customer, but that's not the only option. Whatever works best depends on the nature of the project.



Bruno Lowagie, iText founder